

Master's in Artificial Intelligence for Architecture & Construction

FMP Syllabus

MODULE 10:

Final Master's Project

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MODULE OVERVIEW

The Final Master's Project (FMP) serves as the capstone experience of the master's, requiring students to apply advanced technical skills to a real-life problem or use case from a pre-approved list. This module emphasises a highly innovative and exploratory approach, pushing students to engage with the uncertainties of professional R&D. The FMP is a mandatory collaborative group project (5 students per group) that must be formally defined with clear objectives, timelines, and expected outcomes before starting. Once the topic's suitability is approved, an expert mentor is assigned to guide the team.

Success in the FMP is not judged solely on the successful implementation of the AI solution, as the programme recognises that ambitious projects may face technological limitations. If development is inconclusive, students must redirect their work toward a robust, in-depth analysis of the causes of failure, the lessons learned, and future research directions. Consequently, the final submission can take multiple forms: a successful practical implementation, a viable start-up proposal, or an in-depth academic research paper detailing the project's entire journey and conclusions.

LEARNING OBJECTIVES & COMPETENCIES

The Final Master's Project (FMP) serves as the culminating, integrative module of the master's. It validates that students have successfully achieved the programme's intended learning outcomes by requiring the application of AI-based methods, computational logic and digital technologies to a real-world challenge within the AECO industry.

The FMP promotes project-based, collaborative and research oriented learning, simulating the interdisciplinary workflows of innovation and R&D teams in the built environment. Through this process, students consolidate their technical knowledge and demonstrate the ability to design, implement and critically evaluate AI-driven digital solutions that address technical, operational or sustainability challenges in AECO contexts.

General learning objectives

Upon completion of the FMP, students will be able to:

1. **Synthesise and apply interdisciplinary knowledge**

Acquired throughout the programme to define and address complex, real world AECO problems through the use of artificial intelligence and digital transformation strategies.

2. **Design and execute a structured, research based project plan**
Including clear objectives, milestones, technical methodologies and performance indicators (KPIs).
3. **Develop and validate a functional prototype or data-driven solution**
Integrating AI tools and frameworks within professional AECO workflows (e.g., BIM, Robotics, IoT, Digital Twins etc).
4. **Work collaboratively in multidisciplinary teams**
Demonstrating leadership, adaptability and professional project management using agile or hybrid methodologies.
5. **Evaluate and interpret technical results critically**
Identifying system limitations, data biases and ethical implications in AI-driven decision-making.
6. **Communicate findings effectively**
Through technical documentation, visualizations and professional oral defence to both expert and non-specialist audiences.

Specific technical competencies

By completing the FMP, students will demonstrate advanced proficiency in the following areas:

1. **Interdisciplinary problem-solving and modelling**
 - Formulate and structure complex AECO challenges as data-driven or algorithmic problems.
 - Select and apply appropriate AI techniques (e.g., Machine Learning, Computer Vision, Natural Language Processing, Graph Learning etc.) to generate verifiable outcomes.
2. **Technical integration and prototyping**
 - Integrate AI algorithms into the AECO digital ecosystem and Python-based environments.
 - Design and test interoperable workflows connecting BIM, AR/VR, IoT and Digital Twin systems.
3. **Collaborative project management**
 - Manage distributed teams and complex data structures using collaborative tools (e.g., GitHub, BIM 360, cloud computing environments etc).
 - Apply project management principles (PMI, Agile, Lean Construction) to ensure the effective coordination of scope, time and quality.

4. Critical, ethical and sustainable innovation

- Assess model validity, uncertainty and potential bias in AI systems applied to built-environment data.
- Incorporate responsible AI principles ensuring transparency, data privacy and safety in automated decision-making.
- Evaluate sustainability metrics such as embodied carbon, energy performance and lifecycle impact when relevant.

5. Professional communication and technical documentation

- Prepare a comprehensive technical report and model documentation detailing assumptions, datasets, methods, evaluation metrics and limitations.
- Present and defend results through structured arguments supported by quantitative and visual evidence (e.g., dashboards, performance plots, BIM-linked analytics etc).

Transversal competencies

Throughout the Final Master's Project, students will strengthen key transversal competencies essential for advanced professional and research practice in the digital AECO sector. These competencies promote critical thinking, collaboration, resilience and ethical innovation:

- Strategic and systems thinking
- Team leadership and collaboration
- Resilience and adaptive mindset
- Critical and reflective thinking
- Innovation and research orientation
- Ethical and responsible practice

CALENDAR

Phases of the FMP	Key Session / submissions	Focus / Objective	Dates
Initial phase	Kick-off session	Presentation of the module, objectives, and evaluation. Students' group formation (5 students per group) based on location, affinity, and shared interests.	Wednesday 3 rd December 2025
Qualification & mentor assignment	Proposal submission and mentor assignment	Teams submit their initial proposal (topic, use case, team profiles).	4 th of January 2026
		After reviewing the proposals, the mentors will be assigned.	Week of the 5 th of January
Development Phase: Midway Review	1 st partial delivery	This submission constitutes the problem framing and foundational methodological definition of the FMP.	22 nd of February 2026
	1 st plenary session	Progress Check. Students present a brief of their proposal & progress during the session.	Week of the 2 nd of March 2026
	Groups autonomous work and mentorship sessions	Students continue to develop their projects working collaboratively in their groups. They will have sessions with their mentor.	From beginning of March until beginning of May
	2 nd partial submission	This submission assesses early technical feasibility through a pilot demonstration.	3 rd of May 2026
	2 nd plenary session	Progress check. Students present a brief of their proposal & progress during the session.	Week of the 11 th of May 2026

	Groups autonomous work and mentorship sessions	Students continue to develop their projects working collaboratively in their groups. They will have sessions with their mentor.	From mid-May until beginning of July 2026
	3rd partial submission	This submission represents a comprehensive technical draft of the final project. It acts as a near-complete version of the final report.	12 th of July 2026
Final Phase	3rd plenary	Final preparation. Recap of milestones and a detailed explanation of the defences and their evaluation criteria. The confirmed defence agenda is presented.	Week of the 20 th of July 2026
	Groups autonomous work and mentorship sessions	Students continue to develop their projects working collaboratively in their groups. They will have sessions with their mentor.	From mid-July until the final submission
	Final submission	Students will submit the memory of the project, PPT and any other support material they would use during the presentations.	27 th of September 2026
	Defences	Each group of students will have a day and time to present their project during a public defence. The director and the defence panel will ask questions and give feedback during the session. They will evaluate the projects according to the established criteria.	Weeks of the 5 th and 12 th of October 2026

** In the event that one of the aforementioned deadlines cannot be met, an extension must be requested in writing, indicating the cause or causes that justify it. ZIGURAT's Academic Committee, following a report from the mentor, will approve or deny the extension period requested. Submission after the original deadline or an approved extension deadline may result in a negative evaluation.*

FMP TOPIC AREAS

The Final Master's Project (FMP) must strictly address a real-world challenge or a specific, defined use case within the Architecture, Engineering, Construction, and Operations (AECO) sector, leveraging Artificial Intelligence (AI) technologies.

Students must select their project topic from one or more of the following core thematic areas, ensuring direct relevance to the Master's course content:

- AI applied to BIM technology.
- AI applied to automation & robotics.
- AI applied to Digital Twins.
- AI applied to sustainable design and implementation.
- AI applied to cost estimation and project planning.
- Real-time AI application on the construction site.
- Rehabilitation and AI: structural diagnosis, MEP, and energy efficiency.

Project Scope Requirement

Regardless of the chosen area, the FMP must clearly define:

- A real-life problem and its specific use case.
- The AI model or methodology selected to address the problem.
- The project's potential real-world impact or the critical analysis of the current state-of-the-art that limits the solution.

TEACHING METHODOLOGY

The Final Master's Project (FMP) is delivered through a highly structured, project-based, and self-directed learning methodology, ensuring students take proactive ownership of their innovation process. The core of this methodology is mandatory expert mentorship complemented by structured academic review sessions across three key phases.

Expert mentorship and supervision

Each FMP group works under the guidance of an assigned mentor, whose primary role is to provide strategic guidance, perspective, and encouragement, rather than acting as a technical consultant.

All teams must agree on an operational plan with their mentor, setting the schedule for compulsory mentoring sessions and the milestones for the partial submissions. Students are responsible for proactively contacting their mentor to organise these sessions. Communication between the mentor and the group will take place primarily during the scheduled mentoring sessions. Between sessions, students are expected to work independently on their project, noting down any questions and submitting the materials they have developed at least one week before the next mentoring session.

Role of the mentor

The mentor acts primarily as a guide, strategic advisor, and evaluator throughout the FMP lifecycle. The mentor's function is to provide guidance, perspective, and encouragement, helping students navigate the academic and technical challenges they encounter and make informed decisions as they develop their projects. The mentor monitors the project's adherence to the agreed operational plan and methodology.

Crucially, the mentor does not act as a technical consultant or complete the work for the students. Instead, mentors encourage self-learning, simulate real-world professional scenarios (e.g., assuming a 'client' role), and promote accountability. By reviewing the partial submissions, the mentor ensures the project is on the right track and provides valuable input that contributes to the final assessment through their role in the defence panel.

EDUCATIONAL ACTIVITIES

The FMP module is structured around the following compulsory educational activities, designed to guide students through the complete innovation lifecycle, from problem conceptualisation to professional communication of results.

1. **Proposal:** Introduces the project by defining the real-world problem, the specific use case and the team composition. The proposal formalises the project's initial scope and establishes the foundation for mentor assignment and topic validation.
2. **1st, 2nd, and 3rd partial submissions:** These submissions provide a staged progression through the core phases of the project. They encompass problem framing and contextual analysis, data and model strategy definition, pilot experimentation and feasibility assessment, the evaluation of organisational adoption pathways and the development of a full technical draft based on research, implementation and preliminary results.
3. **Final submission:** Consists of the complete project memory and the defence presentation. This final deliverable articulates the full methodological process, technical outcomes, critical evaluation and the project's relevance to the AECO sector. It serves as the summative academic and professional demonstration of the students' competencies.

Proposal

Deliverables: Do not exceed 1000 words (Arial or Gantari, 11 pt, single spaced).

The project proposal constitutes the formal initiation document for the FMP. It must demonstrate that the team has identified a real and verifiable challenge within the Architecture, Engineering, Construction and Operations (AECO) sector, and that this challenge can be addressed through an AI-enabled methodology.

1.1 Selected topic area

Students must select one or two thematic areas from the topic list. It must align with the programme's technical competencies and demonstrate clear relevance to an AECO workflow.

1.2 Problem statement and use case definition

The problem statement must articulate a precisely delimited AECO challenge, supported by a preliminary analysis of its operational, technical and organisational implications. This section should include:

- A description of the existing workflow or process where the problem occurs.
- The decision-making point(s) affected (e.g., classification, risk prediction, energy optimisation etc).
- Evidence of industry relevance, supported by initial literature, technology review or market observations.
- A specific use case, framed as an applied scenario (e.g., “automated anomaly detection in MEP systems using BMS time-series data”).

1.3 Technical rationale and relevance of AI

Students must justify the suitability of AI for addressing the defined challenge. This includes:

- Identification of potential AI domains (e.g., computer vision, machine learning, natural language processing, reinforcement learning etc).
- Anticipated data types (BIM/IFC, point clouds, time-series, sensor/IoT streams, site imagery etc).
- Expected limitations in current industry solutions or research gaps motivating the proposed investigation.

This rationale must reflect the interdisciplinary problem-solving competencies of the programme.

1.4 Team description

A concise, academically written profile of the 5 team members, highlighting:

- Professional background and relevant domain expertise.
- Technical strengths (e.g., BIM modelling, programming, data analysis, robotics, project management etc).
- Any optional internal organisation or roles agreed upon by the team.

1st partial submission

Deliverables: Do not exceed 1500 words (Arial or Gantari, 11 pt, single spaced)

This submission constitutes the problem framing and foundational methodological definition of the FMP. It must demonstrate analytical depth, methodological coherence and an informed understanding of the AECO context.

1.1 Problem framing analysis

Students must produce a rigorous, evidence-based analysis that includes:

- Market need assessment: A contextualised description of the problem within industry trends such as automation, digital transformation, robotics, sustainability or digital twins etc.
- State-of-Practice review: A concise review of existing tools, methods or academic studies addressing similar challenges.
- Gap identification: A systematic explanation of the limitations, shortcomings or inefficiencies that justify the proposed AI solution.

This section should reflect an academic approach to defining a researchable and impactful AECO problem.

1.2 Data strategy

The data strategy must be technically sound and aligned with recognised industry data practices. It must include:

- Data sources and types: Description of the datasets required (e.g., IFC models, drone imagery, sensor streams, BMS logs, contract documents etc).
- Data preparation workflow: Plans for preprocessing, cleaning, normalisation, feature extraction and annotation.
- Data governance and ethics: Consideration of privacy, security, data bias, representativeness and responsible use of sensitive AECO information.

1.3 AI approach

Students must clearly articulate their intended methodological direction:

- AI model family: Identification of the algorithmic paradigm (e.g., convolutional neural networks, graph neural networks, reinforcement learning agents, hybrid physics-ML models etc).

- Submodels and architectures: Preliminary selection of candidate models and justification (e.g., YOLO for object detection, LSTM for temporal forecasting, BERT for text extraction etc).
- Conceptual pipeline: A high-level description or diagram of how data moves through the system, from input acquisition to output integration, in an AECO workflow.

2nd partial submission

This submission assesses early technical feasibility through a pilot demonstration.

1.1 Pilot demonstration video (Mandatory)

Students must provide a concise video (1-4 minutes) showing:

- Execution of the preliminary model, workflow or prototype.
- Interaction with tools such as Google Collab, Revit/Dynamo, Grasshopper, ROS simulations or IoT dashboards, depending on the chosen topic.
- Representative outputs of the system (e.g., predictions, detections, classifications, control actions etc).

1.2 AI process description (Optional)

If relevant, students may submit a short explanation of:

- The operational logic guiding the AI system.
- The workflow sequence, constraints and assumptions.
- Intermediate calculations or model reasoning (*as applicable*).

1.3 Adoption process analysis (theoretical accepted)

Students must analyse how the proposed solution could be adopted in real AECO organisations:

- Integration pathways: Interoperability with BIM platforms, robotics systems, BMS/IoT infrastructure or enterprise software.
- Change management requirements: Organisational, behavioural and competency implications.
- Scalability and maintenance considerations: Retraining cycles, data refresh mechanisms, system monitoring and long-term governance.

3rd partial submission

This submission represents a comprehensive technical draft of the final project. It acts as a near-complete version of the final report.

1.1 Research and technology review

Students must provide a structured, academically rigorous summary of relevant literature, technologies, datasets and industrial solutions. The review must clarify the specific knowledge gap the project addresses.

1.2 Methods and tools

This section must thoroughly describe:

- Algorithms, model architectures and computational methods.
- Software and environments used (e.g., Revit API, Python, Grasshopper, Colab, PyTorch, TensorFlow, ROS).
- Data analysis procedures, including validation metrics such as precision, recall, F1-score, MAE/MAPE or other discipline-specific measures.

1.3 Development process

Students must document:

- Model iterations and experimental evolution.
- Technical challenges encountered and mitigation strategies.
- Decisions taken during the design and refinement of the workflow.

1.4 Results and discussion

The results should include:

- Quantitative performance indicators, supported by visualisations (confusion matrices, charts, heatmaps, BIM-linked diagrams etc.).
- Interpretation of results in relation to AECO workflows.
- Discussion of implications, limitations and potential improvements.

1.5 References and appendices

- Full APA-compliant reference list.
- Technical appendices including code, datasets, model logs, BIM models, supplementary calculations or external repositories.

Final submission

Consists of: (1) Project Memory and (2) Defence Presentation

1.1 Project memory

Do not to exceed 5000 words (Arial or Gantari, 11 pt, single spaced).

The project memory must be a polished, academically rigorous document that includes:

- Introduction and problem statement.
- Objectives and research questions.
- Methodological framework.
- Development process.
- Results and evaluation.
- Limitations, ethical considerations and risk analysis.
- Contributions to AECO practice and future research directions.
- References and appendices (do not count towards the word limit).

An optional prototype demonstration video may accompany the submission.

1.2 Defence presentation

A professional presentation that summarises:

- Core problem and use case.
- Technical approach and AI model.
- Key results and performance metrics.
- Impact on AECO workflows.
- Conclusions and future work.

The presentation will be evaluated by a panel, which includes the team's mentor, in accordance with the official rubric. Active participation from all group members is mandatory, and the presentation is strictly limited to 10 minutes.

EVALUATION SYSTEM

The Final Master's Project (FMP) assessment culminates in a single, comprehensive final grade. This grade is determined by the final report (70 % of the grade) and defence (30 % of the grade).

Continuous assessment and final grading

While the final grade is singular, the assessment is supported by continuous evaluation of the project process, adherence, and output quality throughout the module:

1. Continuous evaluation by the mentor: The assigned mentor conducts a continuous assessment of the group's progress. This ongoing evaluation focuses on the quality of the partial submissions (problem framing, technical model, pilot project, research analysis), the team's collaborative work, commitment, and compliance with the operational plan. This continuous monitoring is crucial for the mentor to advise the team and contributes to the final assessment through the mentor's input as a panel member.
2. Final submission: The final submission is graded by the mentor.
3. Defence: The panel assesses the defence against the established rubric. The mentor is always part of this panel.

FMP assessment criteria

The final grade is distributed across the following weighted criteria:

Final submission

Criteria	Key assessment focus
1. Problem definition & research context (0-1.75 points)	Depth of research background and literature review, relevance to the built environment and AI, clarity of the problem statement, and identification of gaps or opportunities.
2. Innovation & originality (0-1.75 points)	Creativity in applying AI to construction or architecture, novelty of the idea or approach, and evidence of critical thinking and risk-taking.

3. Technical implementation (0-2 points)	Appropriateness of chosen AI methods/tools, quality of the prototype, simulation, or model, integration with architectural or construction workflows, and handling of data (collection, ethics, etc.).
4. Design thinking & interdisciplinary integration (0-1.75 points)	Consideration of architectural, engineering, and computational aspects, human-centric and sustainable design principles, and usability and real-world applicability.
5. Evaluation & validation (0-1.75 points)	Methodology for testing or validating the solution, use of metrics, benchmarks, or user/tester feedback, and critical reflection on limitations and improvements.
6. Communication & documentation (0-1 point)	Clarity and structure of the final report, visual communication (diagrams, charts, etc.), and proper referencing and adherence to academic integrity.

Defence

Presentation & professionalism (0-1.5 points)	Quality of the final presentation (visual or demo etc.), student's ability to answer questions and defend their work, and engagement with feedback and critique.
Core problem and use case (0-2 points)	Relevance of the identified industry pain point, the clarity of the target user, and the strength of the proposed value proposition.
Technical approach and AI model (0-2 points)	Rigour of the data strategy, the justification for the chosen AI methodologies, and the logic of the technical architecture.

Key results and performance metrics (0-1.5 points)	Evidence of success through selected performance metrics, accurate analysis, and effective data visualisation.
Conclusions and future work (0-1.5 points)	Reflection of the team on their achievements and limitations, as well as the feasibility of their roadmap for scaling.
Teamwork (0-1.5 points)	Equitable distribution of speaking roles, and the collaborative support displayed during the defence.

APPENDIX

Description of each of the topics

AI applied to BIM technology

Use AI to enrich, validate and generate BIM data, so models become machine-checkable, interoperable and more useful across the project lifecycle.

- IFC/Revit models – Auto classify elements (e.g., walls vs. curtain walls) and add missing parameters using graph learning or NLP on element names/families etc.
- AI-assisted clash & constructability checking - Learn from historical clash reports to predict “high-risk” clashes and propose automated resolutions ranked by constructability.
- Automated code & standards compliance - Use NLP + rule learning to map building codes to model queries (e.g., egress widths), then flag noncompliance with traceable explanations.
- 2D to 3D BIM reconstruction - Convert plan PDFs and point clouds into parametric elements with CV + procedural modelling; estimate confidence for designer review.
- Model version anomaly detection - Detect unintended geometry/parameter changes between versions (BIM diff) with graph-based change detection and alerting.

AI applied to automation & robotics

Combine perception, planning and control so robots can navigate, manipulate and assist in construction/assembly tasks safely.

- Vision-based material handling - Detect pallets, rebar and components; estimate pose for grasping/placement using object detection + 6D pose networks.
- Task & path planning with RL - Learn task sequences and collision-free paths in partially known sites; integrate cost maps from BIM.
- Construction-site SLAM - Robust mapping in dust/occlusion with sensor fusion (LiDAR + RGB-D) and loop-closure tuned to repetitive structures.
- Human-robot collaboration safety - Predict human trajectories and enforce dynamic safety zones; evaluate with near-miss metrics.
- Multi-robot coordination - Schedule and deconflict multiple mobile robots (e.g., delivery + cleaning) using multi-agent RL or auction algorithms.

AI applied to Digital Twins

Build “live” twins that fuse BIM/GIS with sensors to predict, simulate and optimize building or campus performance.

- Sensor fusion & state estimation - Infer hidden states (occupancy, thermal mass) from

IoT streams with Kalman/particle filters and learned priors from BIM.

- Predictive maintenance (HVAC/elevators) - Remaining-useful-life models on vibration, current or temperature; schedule interventions to minimize downtime.
- Structural health in the twin - Update stiffness/damage parameters from strain/acceleration data; visualize in a 3D twin for rapid triage.
- Occupant-centric control - Learn comfort/IEQ preferences and drive setpoints (temperature/ventilation/lighting) with safe RL under constraints.
- What-if simulation & optimization - Run fast surrogates (emulators) for energy/daylight/airflow to test retrofits and rank by ROI + carbon impact.

AI applied to sustainable design and implementation

Use AI + simulation to cut energy, carbon and resource use from building to district scale while safeguarding comfort and equity.

- Operational energy prediction - Hybrid physics-ML models to predict EUI from early-stage geometry and climate; quantify uncertainty for design decisions.
- Daylight/solar optimization - Optimize façade geometry/shading with multi-objective search (sDA/ASE vs. cooling loads).
- Embodied carbon estimation - Map materials in BIM to EPD/LCA databases with NLP; estimate A1–A5 (EN 15978) impacts and propose low-carbon alternates.
- Adaptive controls for efficiency - Learn schedules for HVAC/lighting based on occupancy forecasts; evaluate comfort and peak-load reduction.
- Smart-city mobility & microclimate - Predict pedestrian flows/heat-island hotspots; recommend trees/canopies or routing that balances comfort and access.

AI applied to cost estimation and project planning

Turn heterogeneous project information (BIM, schedules, contracts, RFIs) into reliable cost/schedule forecasts and automated documentation flows.

- 5D cost takeoff from BIM - Extract quantities and map to cost codes; learn cost multipliers from historical bids and site conditions.
- Probabilistic schedule forecasting - Predict task durations from features (crew mix, weather, rework) and produce risk-aware schedules with Monte Carlo simulation.
- Change-order impact analysis - Model downstream cost/schedule impacts of design changes; surface the top drivers with SHAP-style explanations.
- Document intelligence (OCR/NLP) - Extract clauses, milestones and submittal requirements from specs/contracts; auto-link to model elements and WBS.
- Automated submittal/RFI routing - Classify and route to responsible parties; generate draft responses using retrieval-augmented generation over prior projects.

Real-time AI application on the construction site

Stream and analyze imagery/IoT to monitor progress, quality and safety, closing the loop to 4D plans in near-real time.

- Progress tracking vs. 4D - Align drone or 360° video point clouds to BIM; auto-percent-complete per activity with confidence and variance to plan.
- Safety & PPE compliance - Detect PPE usage, proximity to hazards, and unsafe postures with on-edge inference; trigger real-time alerts.
- Material/asset tracking - Fuse RFID/UWB/BLE with vision to locate materials and prevent loss; forecast stock-outs.
- On-site QA/QC with AR - Compare as-built scans to tolerances; highlight deviations through AR overlays for installers.
- Weather & micro-risk nowcasting - Predict short-term wind/rain for crane lifts and concrete pours; auto-adjust schedules and method statements.

Rehabilitation and AI: structural diagnosis, MEP and energy efficiency

Diagnose defects and prioritize interventions for existing buildings to extend life, cut energy use and improve resilience.

- Surface damage detection - Use CV on photogrammetry/thermal images to map cracks, spalling, moisture; estimate severity and progression.
- Vibration-based damage localization - Identify stiffness loss via modal analysis and sparse Bayesian learning; validate on lab/field data.
- MEP anomaly detection - Detect faults in AHUs/chillers/pumps and duct leakage from BMS time-series with interpretable models.
- Retrofit recommendation engine - Optimize retrofit bundles (insulation, glazing, heat pumps) against cost, payback, and carbon; produce staged roadmaps.
- Energy audits from smart meters (NILM) - Disaggregate loads to discover waste and schedule operational fixes before capex retrofits.

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